KAIZEN MATHS

Lesson starter

Discrete Random Variables and Probability Distributions

Expectation & Variance

Name _____

S1

Calculate E(X) for:

x	2	3	4	5
P(X=x)	0.31	0.38	0.06	0.25

S2

Calculate Var(X) for:

x	2	3	4	5	6
P(X=x)	0.26	0.16	0.37	0.16	0.05

S3

If E(X) = 5, find E(1 - 3X)

S4

Find E(X), Var(X), and SD(X):

x	0	1	2	3
P(X=x)	0.10	0.50	0.30	0.10

Show your working

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Exit ticket

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E1

Calculate E(X) for:

x	0	1	2	3
P(X=x)	0.31	0.06	0.13	0.50

E2

Calculate Var(X) for:

x	1	2	3	4	5
P(X=x)	0.20	0.25	0.05	0.10	0.40

E3

If E(X) = 4, find E(1 - 5X)

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Exit ticket

Discrete Random Variables and Probability Distributions

Expectation & Variance

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Calculate E(X) for:

x	0	1	2	3
P(X=x)	0.31	0.06	0.13	0.50

E2

Calculate Var(X) for:

x	1	2	3	4	5
P(X=x)	0.20	0.25	0.05	0.10	0.40

E3

If E(X) = 4, find E(1 - 5X)

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Discrete Random Variables and Probability Distributions

Expectation & Variance

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P(X=x)	0.31	0.06	0.13	0.50

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P(X=x)	0.20	0.25	0.05	0.10	0.40

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If E(X) = 4, find E(1 - 5X)

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x	1	2	3	4	5
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E3

If E(X) = 4, find E(1 - 5X)

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Teacher answers & working

Discrete Random Variables and Probability Distributions · Expectation & Variance

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S1

Calculate E(X) for:

x	2	3	4	5
P(X=x)	0.31	0.38	0.06	0.25

Answer:

3.25

Working:

Use $E(X) = \text{sum } xP(X=x)$

$E(X) = 2(0.31) + 3(0.38) + 4(0.06) + 5(0.25)$

$= 0.62 + 1.14 + 0.24 + 1.25$

$E(X) = 3.25$

S2

Calculate Var(X) for:

x	2	3	4	5	6
P(X=x)	0.26	0.16	0.37	0.16	0.05

Answer:

1.38

Working:

Calculate E(X)

$E(X) = 2(0.26) + 3(0.16) + 4(0.37) + 5(0.16) + 6(0.05) = 3.58$

Calculate $E(X^2)$

$E(X^2) = 2^2(0.26) + 3^2(0.16) + 4^2(0.37) + 5^2(0.16) + 6^2(0.05) = 14.20$

Use $Var(X) = E(X^2) - [E(X)]^2$

$Var(X) = 14.20 - (3.58)^2 = 1.38$

1.38

Teacher answers & working

Discrete Random Variables and Probability Distributions · Expectation & Variance

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S3

If E(X) = 5, find E(1 - 3X)

Answer:

-14

Working:

Use linearity of expectation

$E(aX + b) = aE(X) + b$

Substitute

$1 - 3E(X) = 1 - 3(5)$

-14

S4

Find E(X), Var(X), and SD(X):

x	0	1	2	3
P(X=x)	0.10	0.50	0.30	0.10

Answer:

$E(X)=1.40$, $Var(X)=0.64$, $SD=0.80$

Working:

Calculate E(X)

$E(X) = 0(0.10) + 1(0.50) + 2(0.30) + 3(0.10) = 1.40$

Calculate $E(X^2)$

$E(X^2) = 0^2(0.10) + 1^2(0.50) + 2^2(0.30) + 3^2(0.10) = 2.60$

Calculate variance and SD

$Var(X) = 2.60 - (1.40)^2 = 0.64$

SD(X) = sqrt(0.64) = 0.80

$E(X)=1.40$, $Var(X)=0.64$, $SD=0.80$

Teacher answers & working

Discrete Random Variables and Probability Distributions · Expectation & Variance

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E1

Calculate E(X) for:

x	0	1	2	3
P(X=x)	0.31	0.06	0.13	0.50

Answer:

1.82

Working:

Use $E(X) = \text{sum } xP(X=x)$

$E(X) = 0(0.31) + 1(0.06) + 2(0.13) + 3(0.50)$

$= 0.00 + 0.06 + 0.26 + 1.50$

$E(X) = 1.82$

E2

Calculate Var(X) for:

x	1	2	3	4	5
P(X=x)	0.20	0.25	0.05	0.10	0.40

Answer:

2.69

Working:

Calculate E(X)

$E(X) = 1(0.20) + 2(0.25) + 3(0.05) + 4(0.10) + 5(0.40) = 3.25$

Calculate $E(X^2)$

$E(X^2) = 1^2(0.20) + 2^2(0.25) + 3^2(0.05) + 4^2(0.10) + 5^2(0.40) = 13.25$

Use $Var(X) = E(X^2) - [E(X)]^2$

$Var(X) = 13.25 - (3.25)^2 = 2.69$

2.69

Teacher answers & working

Discrete Random Variables and Probability Distributions · Expectation & Variance

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E3

If E(X) = 4, find E(1 - 5X)

Answer:

-19

Working:

Use linearity of expectation

$E(aX + b) = aE(X) + b$

Substitute

$1 - 5E(X) = 1 - 5(4)$

-19